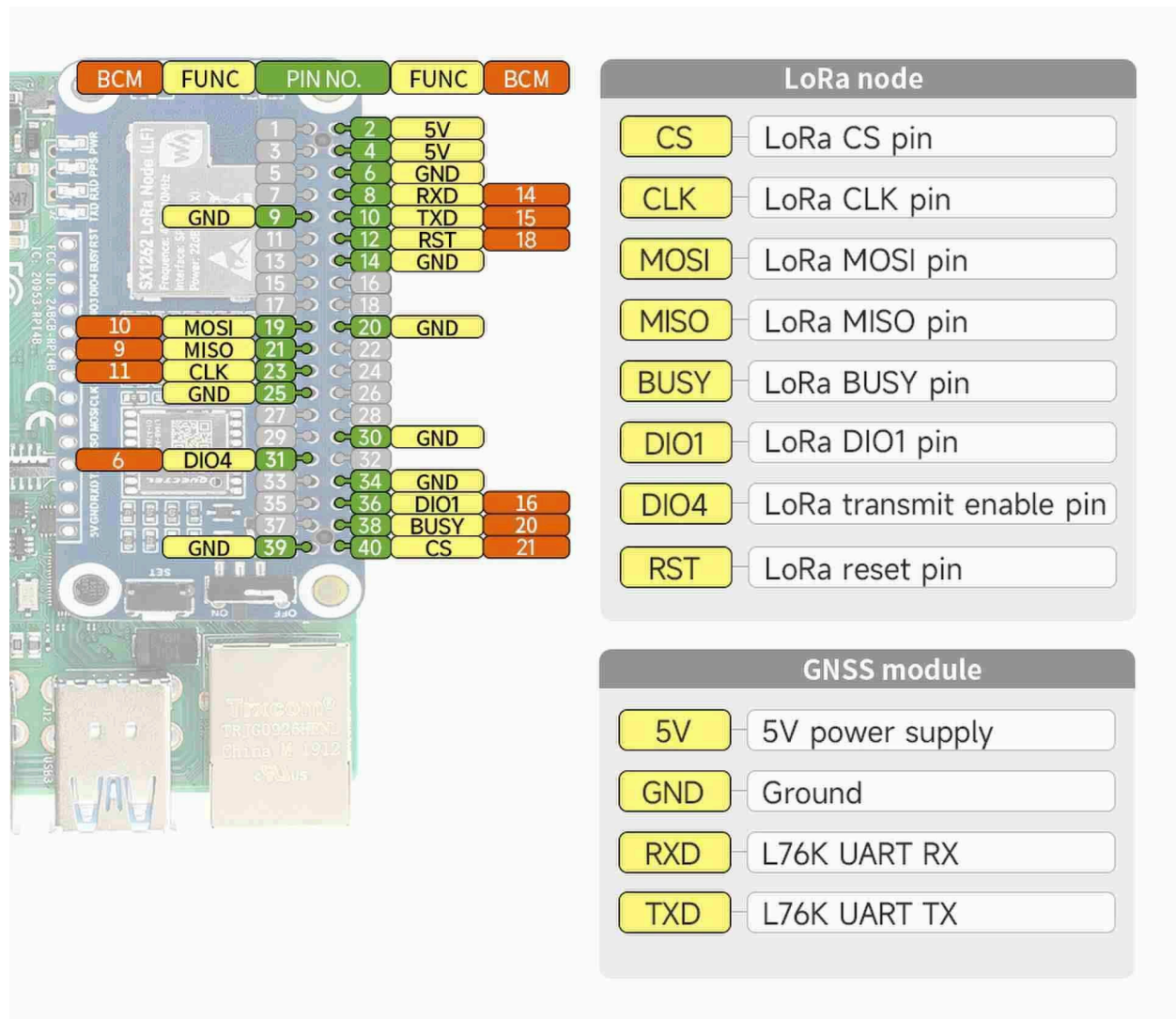
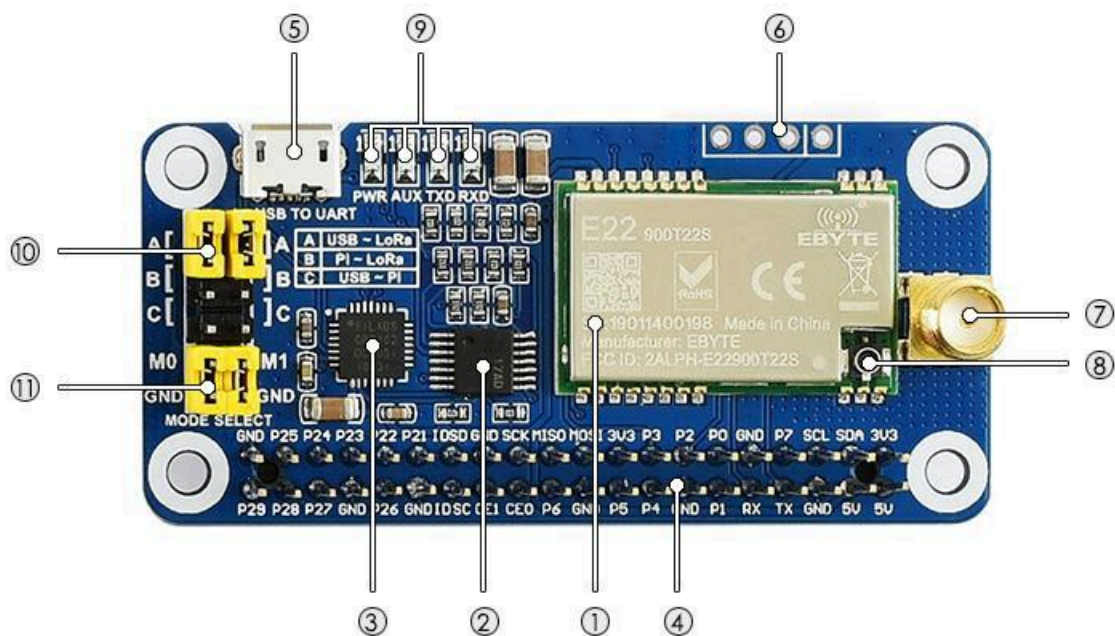
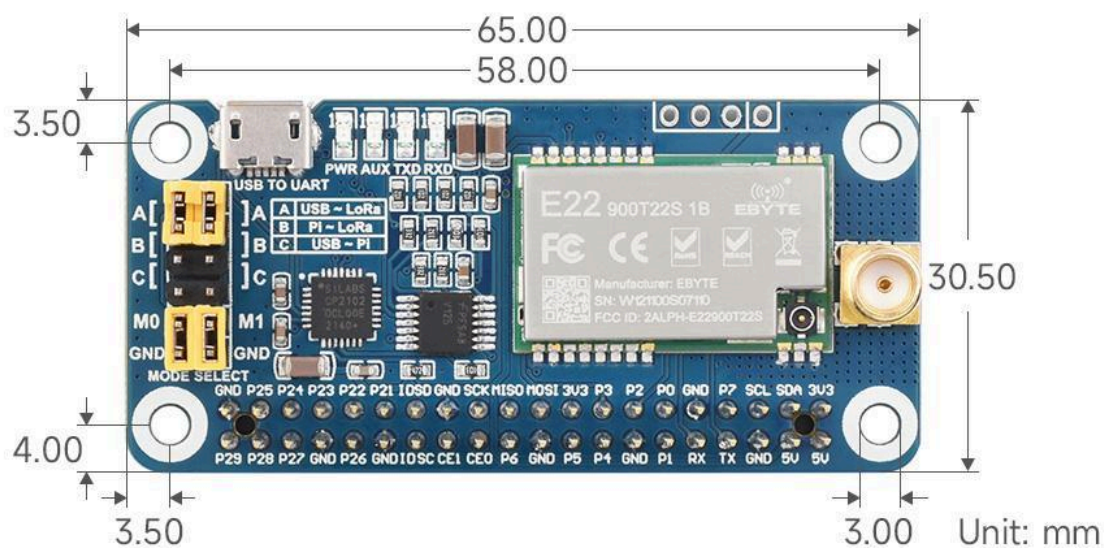


Raspberry Pi 5 ↔ SX126X LoRa HAT Connection

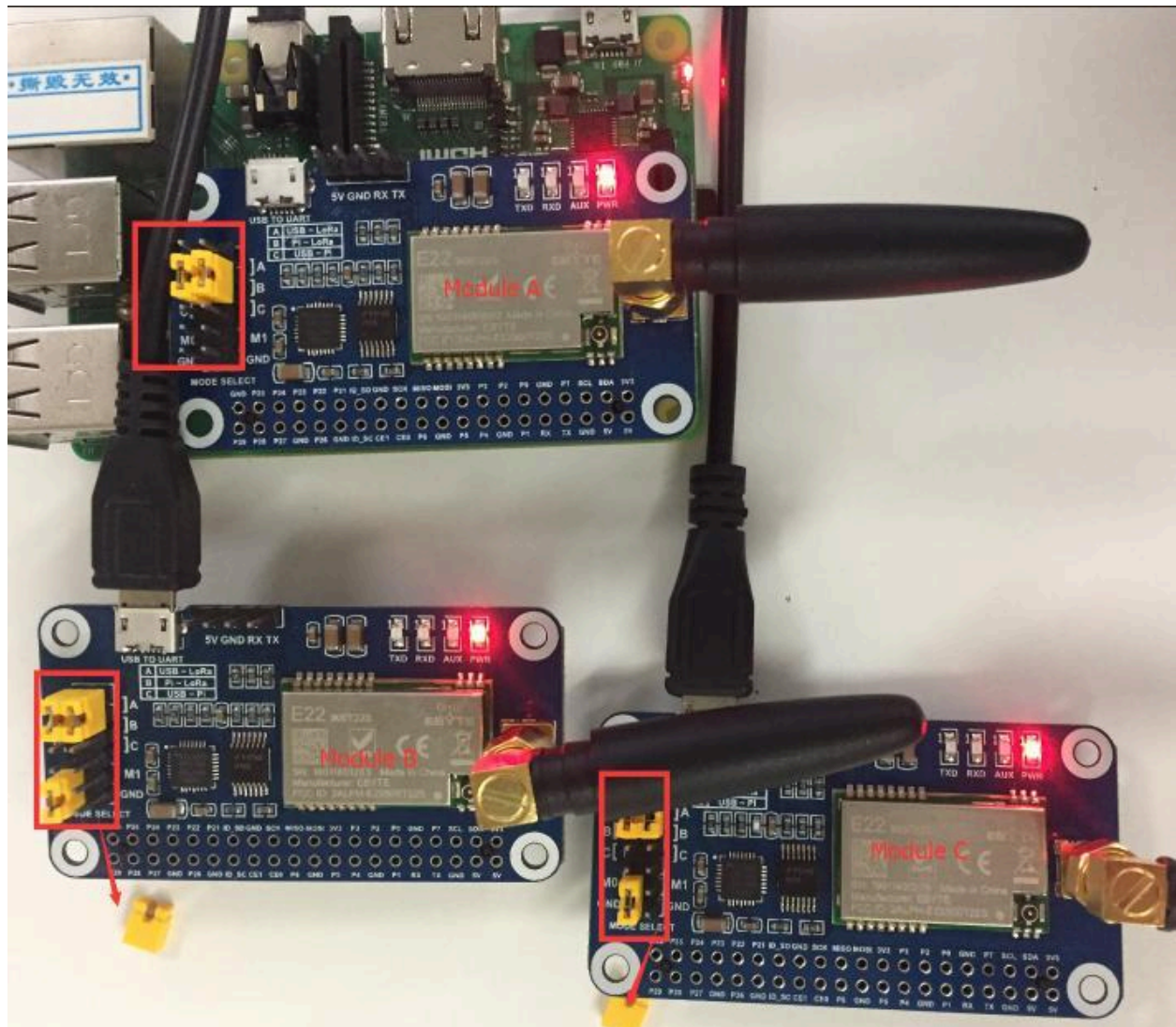
Use SPI connection between the Raspberry Pi 5 and SX126X LoRa HAT.

Connection Diagram







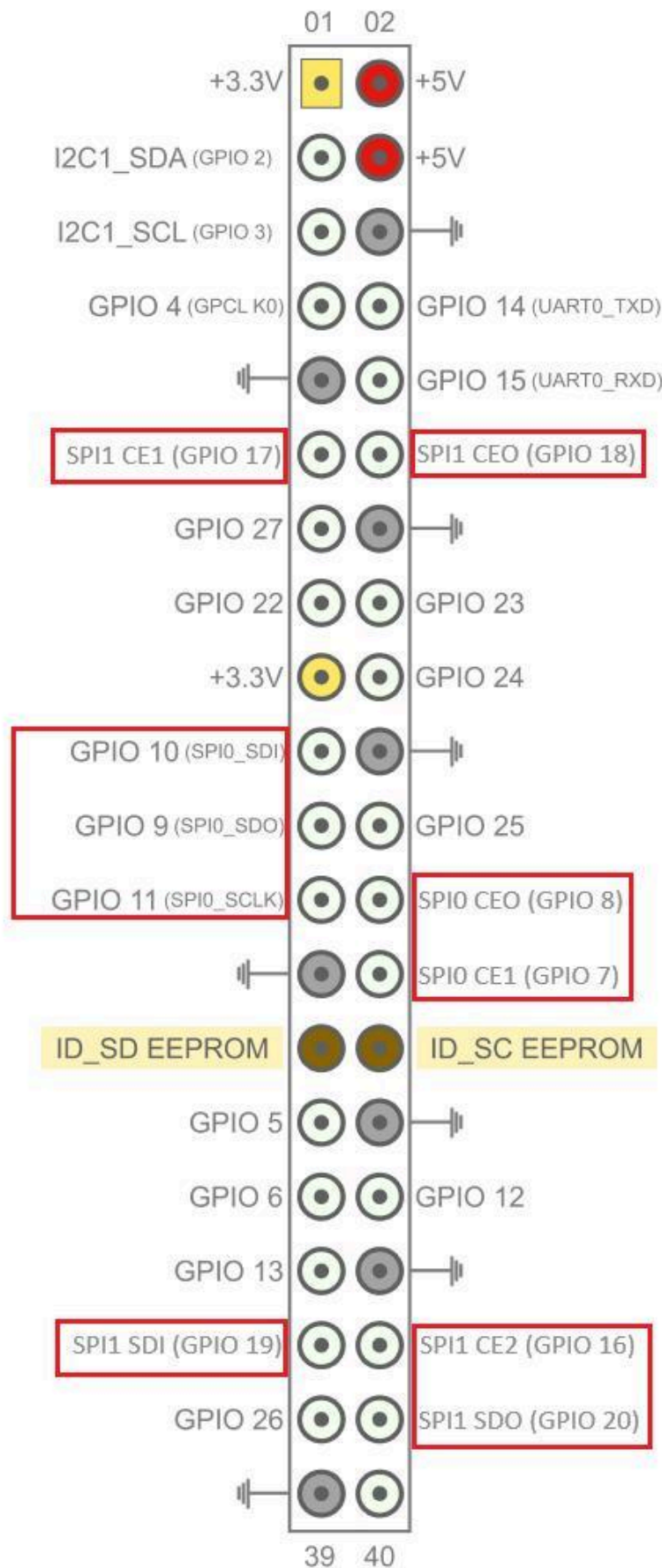


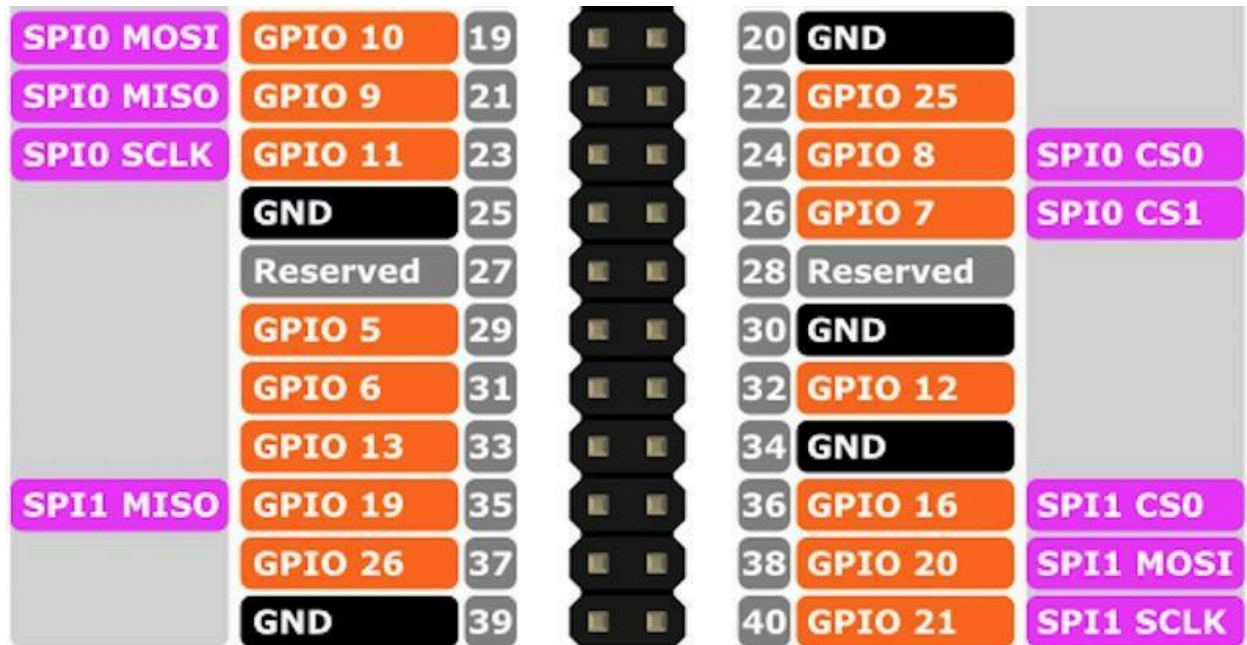
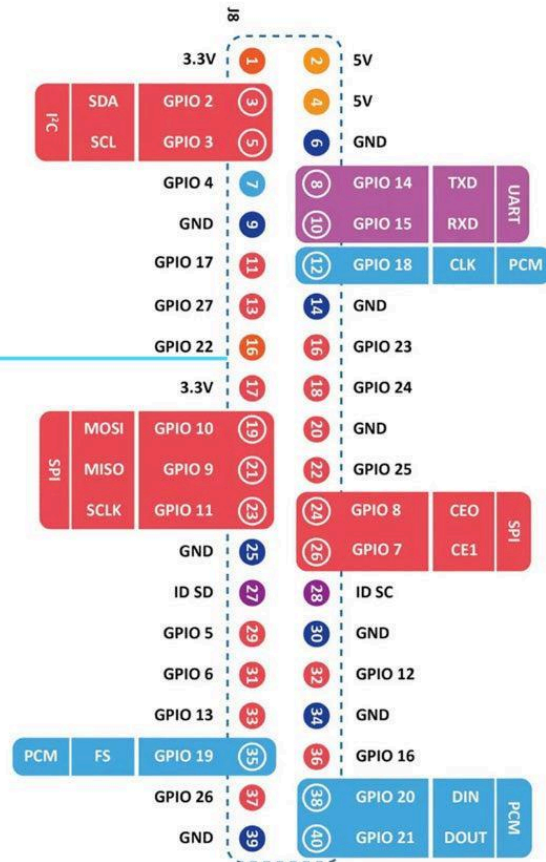
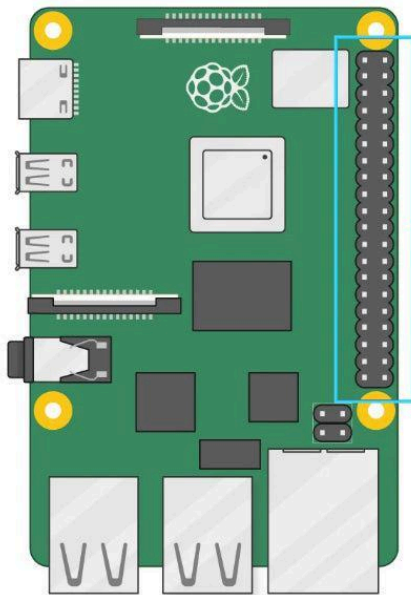
GPIO Wiring Table

SX126X LoRa HAT	Raspberry Pi 5 GPIO	Pin Number
VCC	3.3V	Pin 1
GND	GND	Pin 6
MISO	GPIO9 (MISO)	Pin 21
MOSI	GPIO10 (MOSI)	Pin 19
SCK	GPIO11 (SCLK)	Pin 23
NSS / CS	GPIO8 (CE0)	Pin 24

RESET	GPIO22	Pin 15
BUSY	GPIO24	Pin 18
DIO1	GPIO25	Pin 22

Raspberry Pi 5 GPIO Layout







GPIO Pinout – Rasp Pi 1 Model B+/Rasp Pi 2 Model B

40-pin GPIO Image from element14.com



www.raspberry-pi-geek.com/howto/gpio-pinout-rasp-pi-1-model-b-rasp-pi-2-model-b

Pin#	NAME	NAME	Pin#
01	3.3v DC Power	DC Power 5v	02
03	GPIO02 (SDA1, I2C)	DC Power 5v	04
05	GPIO03 (SCL1, I2C)	Ground	06
07	GPIO04 (GPIO_GCLK)	(TXD0) GPIO14	08
09	Ground	(RXD0) GPIO15	10
11	GPIO17 (GPIO_GEN0)	(GPIO_GEN1) GPIO18	12
13	GPIO27 (GPIO_GEN2)	Ground	14
15	GPIO22 (GPIO_GEN3)	(GPIO_GEN4) GPIO23	16
17	3.3v DC Power	(GPIO_GEN5) GPIO24	18
19	GPIO10 (SPI_MISO)	Ground	20
21	GPIO09 (SPI_MISO)	(GPIO_GEN6) GPIO25	22
23	GPIO11 (SPI_CLK)	(SPI_CE0_N) GPIO08	24
25	Ground	(SPI_CE1_N) GPIO07	26
27	ID_SD (I2C ID EEPROM)	(I2C ID EEPROM) ID_SC	28
29	GPIO05	Ground	30
31	GPIO06	GPIO12	32
33	GPIO13 PWM 1	Ground	34
35	* GPIO19 SPI 1 MISO	SPI 1 CE 2 GPIO16	36
37	GPIO26	SPI MOSI GPIO20	38
39	Ground	SPI 1 SCLK GPIO21	40

* GPIO 19 = PWM 1

```
1. import spidev
2. spi = spidev.SpiDev()
3. spi.open(0, CHIP_SELECT_0_OR_1)
4. spi.max_speed_hz = 1000000
5. spi.xfer([value_8bit])
```

I2C/SPI/UART/PWM pinout v0.1 ttfong01 2019may15

GPIO Pin	PWM0/PWM1
GPIO12	PWM0
GPIO18	PWM0
GPIO13	PWM1
GPIO19	PWM1

SPI - Serial Peripheral Interface

SPI0 pins in BCM mode are: 9, 10, 11 + 7/8

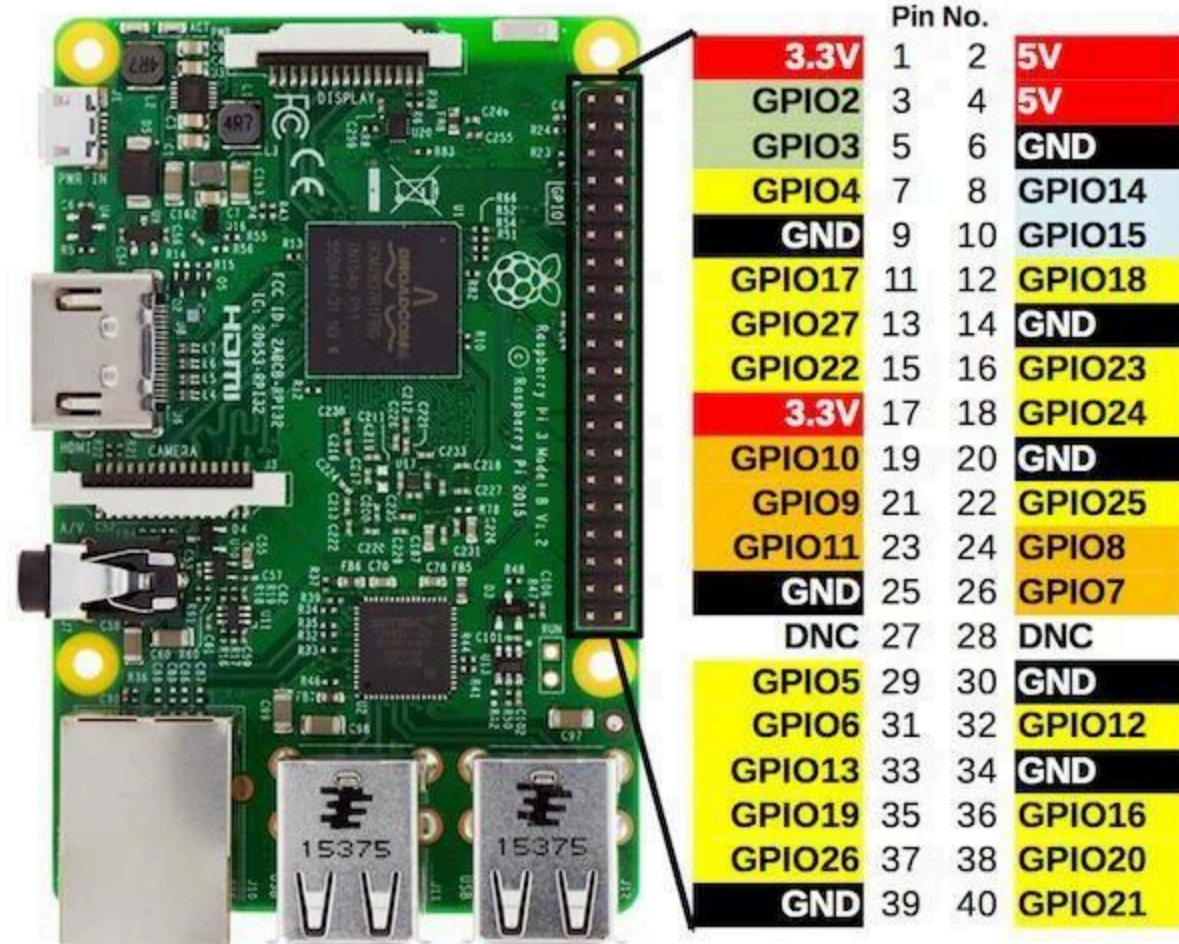
<https://pinout.xyz/pinout/spi>



Legend

- GPIO (General Purpose IO)
- SPI (Serial Peripheral Interface)
- I2C (Inter-Integrated Circuit)
- UART (Universal Asynchronous Receiver/Transmitter)

- Ground
- 5V (Power)
- 3V (Power)



Enable SPI on Raspberry Pi OS

Open terminal:

```
sudo raspi-config
```

Go to:

Interface Options → SPI → Enable

Reboot:

```
sudo reboot
```

Check SPI

```
ls /dev/spi*
```

You should see:

```
/dev/spidev0.0  
/dev/spidev0.1
```

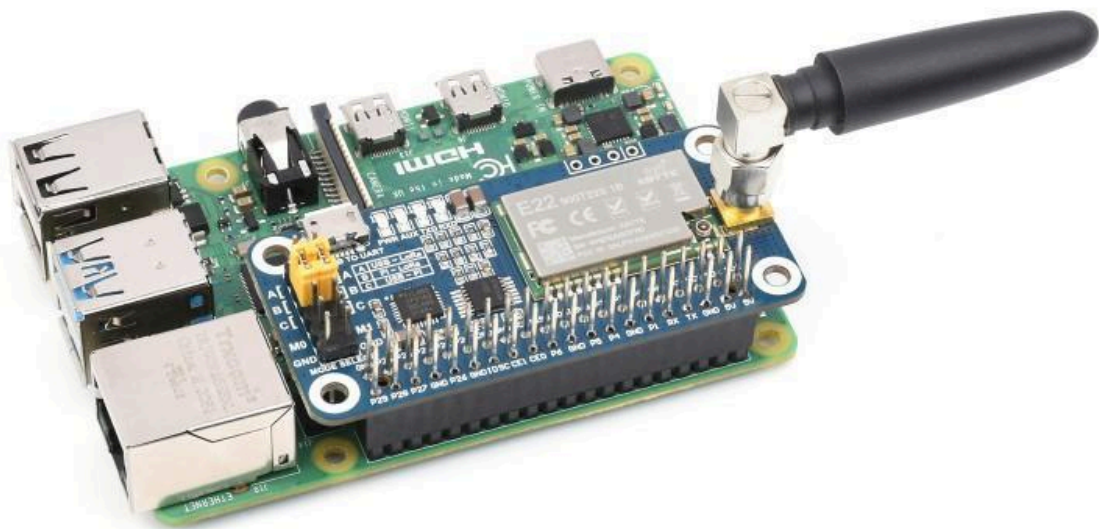
Install Python Library

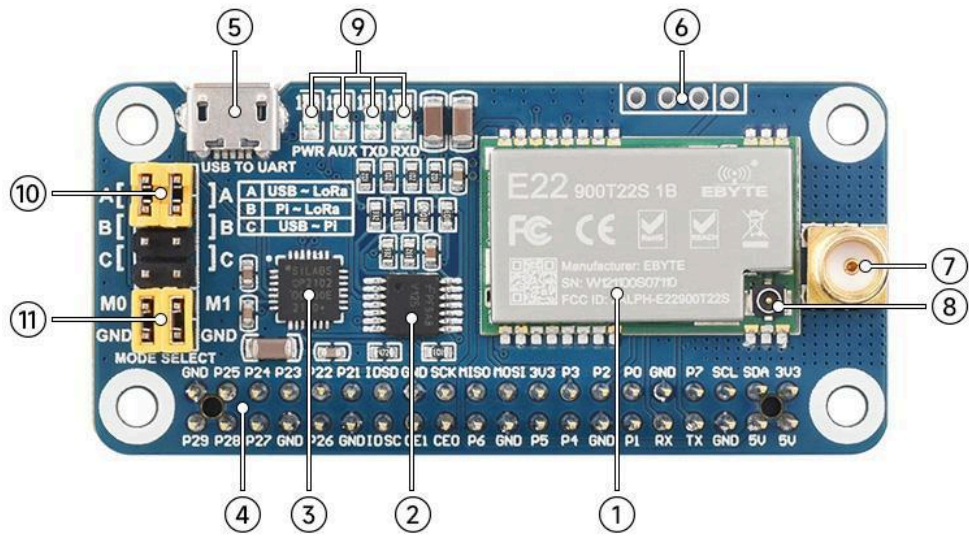
```
sudo apt update  
sudo apt install python3-pip -y  
pip install spidev gpiozero
```

Basic Test Code

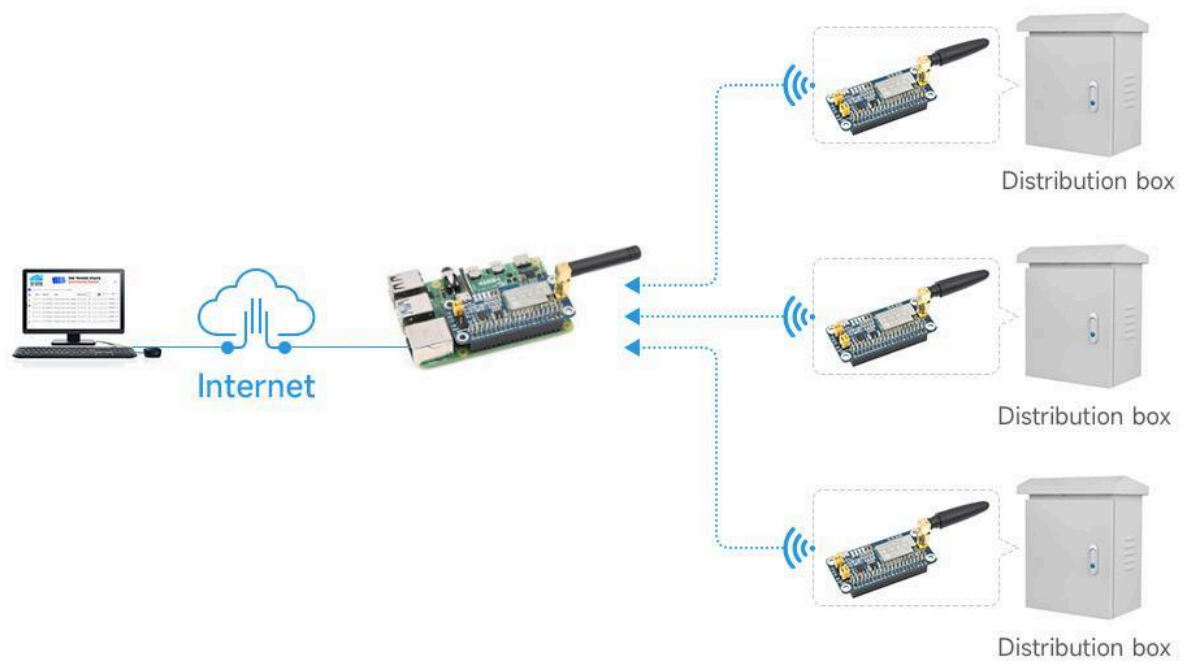
```
import spidev  
import time  
  
spi = spidev.SpiDev()  
spi.open(0, 0)  
spi.max_speed_hz = 1000000  
  
print("SPI Connected Successfully")  
  
while True:  
    time.sleep(1)  
    print("LoRa Ready")
```

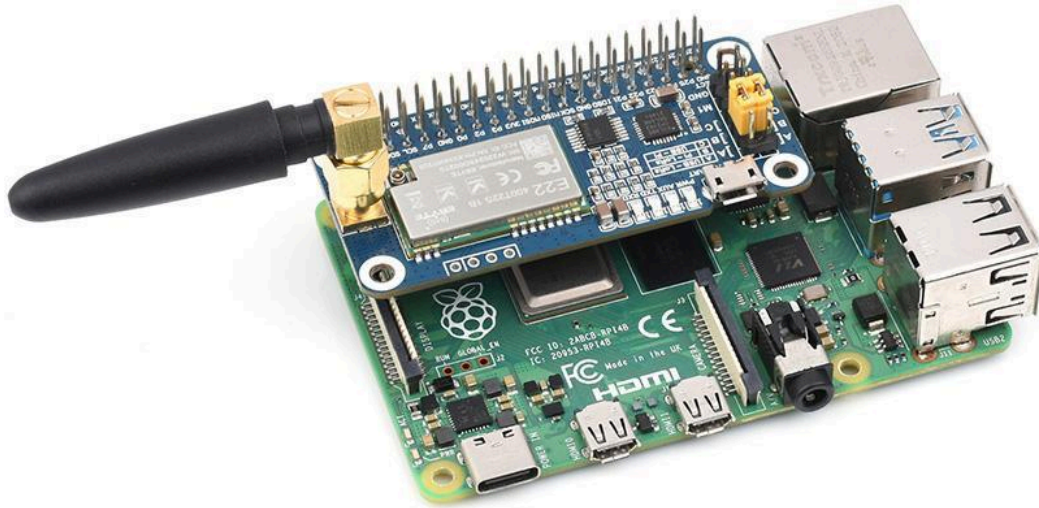
SX126X Module Example











Important

- SX126X works on **3.3V only**
- Do NOT connect to 5V
- Enable SPI before testing
- Raspberry Pi 5 supports SPI very well for LoRa communication

Official documentation:

- [Raspberry Pi GPIO Documentation](#)
- [Waveshare SX126X Wiki](#)